

Abhinav Kumar

Senior Backend Engineer | Python & FastAPI | Microservices & Kubernetes | AI/LLM Systems
Bengaluru, India | abhinav6400625@gmail.com | +91-9916118034
LinkedIn | GitHub | Portfolio

Professional Summary

Senior Backend Engineer with 5+ years designing, building, and operating Python microservices with FastAPI in containerized, cloud-native environments (Docker, Kubernetes, AWS). Led a 6x scale-up of a customer-enrichment platform (30K+ records/day), cost work that saved \$300K, and a self-service API adopted by all 12 tenants. Built production AI systems, including Doc Intent, a LangGraph and RAG-based design-intent engine serving **1K+ requests/day**, and LLM classification at 96% accuracy across 30K–50K calls/day. Strong in testing (Pytest, Postman), API security (authentication and authorization), performance and scalability, CI/CD, production support (PagerDuty, New Relic), and documentation. Mentor engineers and partner closely with product, data, QA, and DevOps teams.

Technical Skills

Languages:	Python (AsyncIO, ThreadPoolExecutor), SQL, TypeScript
Backend & APIs:	FastAPI, Flask, REST APIs, Microservices, OpenAPI/Swagger, Postman, JWT, OAuth2, Rate Limiting
Testing:	Pytest, Integration & Contract Testing, Load Testing, TestRail
DevOps & Cloud:	Kubernetes, Docker, Terraform, GitLab CI/CD, Git; AWS (S3, Lambda, Step Functions, API Gateway, Fargate, ECS, EC2, DynamoDB, ECR, VPC)
Data & Caching:	Redis, DynamoDB, Snowflake, dbt, OpenSearch, PostgreSQL, Data Modeling, Advanced SQL
AI / LLM:	LangGraph, AI Agents, RAG, Embeddings, Vector Search, OpenAI, Anthropic, Grok (xAI), Ollama, HuggingFace, Snowflake Cortex, Prompt Engineering, LLM Evaluation
AI Dev Tools:	Claude Code, Cursor, Google Antigravity, AI-Assisted Development
Observability:	New Relic, PagerDuty, CloudWatch, Grafana, Prometheus, Logging, Tracing, Alerting
Practices:	Distributed Systems, Event-Driven Architecture, SOLID, Design Patterns, Code Reviews, Agile

Experience

Cimpress Technology (Vistaprint), MAT – Marketing Automation Technology	Bengaluru, India
<i>Senior Software Engineer</i>	<i>Jul 2024 – Present</i>

- **Led – Doc Intent (AI Design-Intent Engine):** Built Doc Intent, an AI product where users upload a design image (flyer, card, signage) and the system infers its intent, including industry, cause or occasion, product type, and other attributes used for mass customization. Orchestrated the multi-step reasoning as a **LangGraph** agent workflow and grounded predictions with **RAG** over embeddings of the taxonomy; serves **1K+ requests/day** in production.
- **Led – Enrichment Platform Scale-Up:** Scaled customer-enrichment infrastructure from about 4–5K to **30K+ records/day** by redesigning parallel execution across AWS Lambda, Step Functions, Fargate, EC2, AsyncIO, and ThreadPoolExecutor, supporting workloads up to a 1M-record requirement.
- **Led – API Cost Optimization:** Identified redundant Google Places and SERP API usage and designed SERP-result analysis to skip unnecessary Places requests, saving about **\$300K**.
- **Led – Self-Service Enrichment API:** Designed and launched a self-service API (API Gateway, Lambda, Step Functions) that replaced CSV-based workflows and manual support; adopted by **all 12 tenants**. Added request-boundary validation (company, website/email, country, domain, duplicates, malformed input) to keep bad data out of downstream workflows.
- **Led – AI Industry Classification:** Built LLM-based classification to replace brittle SERP-to-taxonomy mappings, mapping companies to 102 categories at about **96% manually validated accuracy**. Added a 90% confidence threshold and a Slack review loop for uncertain cases; handles **30K–50K LLM calls/day**.
- Led the redesign of the Organization ID architecture, establishing a consistent organization-level grouping model across enrichment workflows.
- Re-engineered address extraction with LLM-based field classification instead of format-specific rules, raising coverage by about 10% and accuracy by 5% and eliminating observed extraction failures.
- Moved data-quality checks from field-level rules to semantic validation using Snowflake rules, Snowflake Cortex, and a **Grok-powered bot** that runs LLM-based data-quality checks across a large set of parameters, flagging suspicious records before tenant delivery. Designed Snowflake schemas and dbt pipelines for 3–4 tenants on datasets of about 23M records.
- Built and maintain 20+ REST APIs with FastAPI and Flask on DynamoDB, Snowflake, and OpenSearch, with async processing, JWT authentication and authorization, and OpenAPI documentation; build and debug APIs

with Postman.

- Improved performance and scalability: a DynamoDB caching layer that cut pipeline AWS costs by about **20–30%**, OpenSearch relevance tuning for the EVO Portal, and load testing that caught concurrency bottlenecks before production rollout.
- Built CI/CD and infrastructure automation with GitLab, Docker, and Terraform; deploy services on Kubernetes and ECS with monitoring in New Relic and CloudWatch. Write unit and integration tests with Pytest, including automated Lambda integration tests in CI, and track release sign-off with QA in TestRail.
- Own services end-to-end, from requirements and design through deployment and PagerDuty on-call; use AI-assisted development (**Cursor**, Claude Code) to speed up large-scale feature development and refactoring. Partner with product and data-engineering stakeholders, mentor 2 junior engineers, and review PRs daily.

Cimpress Technology

Bengaluru, India

Software Engineer

Aug 2021 – Jul 2024

- Built and maintained Python services and enrichment pipelines for high-volume customer data, including production monitoring, incident resolution, and optimization.
- Integrated GPT-based classification for industry mapping and address traits, reaching about 90% accuracy in early validation and reducing manual classification effort.
- Converted Lambda integrations into API-based workflows to enable automated integration testing, reducing failure scenarios and increasing deployment confidence.
- Revamped Organization ID functionality from scratch, working with Printdeal to resolve organization identity issues.
- Implemented website caching (Redis) to cut repeated external requests, and worked on GCP cost optimization while preserving enrichment accuracy.

Projects

LLM Duel Arena – Open Source

GitHub

- Built a full-stack platform where LLMs compete in games such as Chess, Tic-Tac-Toe, Racing, and Word Association.
- Designed a unified provider abstraction for OpenAI, Anthropic, Ollama, and HuggingFace, so game logic stays provider-independent and new models, providers, and games plug in without changing the core engine.
- Implemented game-specific prompt templates, constrained generation, move validation, and result tracking, reaching 95%+ legal move generation.

Education

Siddaganga Institute of Technology, Tumakuru

2017 – 2021

B.E., Computer Science & Engineering | CGPA: 8.4 / 10